The Levitron Lab: An Interactive Curriculum for Magnetic Levitation Physics

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We describe *The Levitron Lab*, a browser-based collection of interactive simulations that teach how magnetic levitation toys evade Earnshaw's theorem by climbing a dimensionality ladder from a static saddle-point primer to full rigid-body and driven-rotor dynamics. Each of five modules foregrounds a live, animated levitating body with sliders on every physical parameter, diagnostic plots, and inline equations. Feedback modules implement full PID control with live Jacobian eigenvalues and time-series traces of position and coil drive; the gyroscopic module implements the symmetric-top nutation analysis from the author's MIT 8.223 project. The application is open source, runs entirely client-side, and can be launched in one click via GitHub Codespaces or deployed as a public website on Railway. Supplementary interactive materials, figure-capture scripts, and this manuscript are available at https://github.com/saleemaldajani/levitron-lab.

I. INTRODUCTION

Earnshaw's theorem forbids stable equilibrium for a magnetic dipole in a static field: wherever forces balance, the potential is a saddle[3]. Yet levitating globes, spinning tops, and motor-driven floaters are sold as toys. Each evades the theorem by a distinct mechanism: time-varying feedback fields; gyroscopic spin; or frequency-locked rotation in a driven field[4–6].

The Levitron Lab presents all three escape routes in one curriculum, ordered by *which directions are unstable*—a classification articulated in conversation with an experimental atomic physicist who keeps a classic Levitron on his desk as a tabletop model of a magnetic atom trap[10]. The software is intended for upper-level mechanics and E&M courses, outreach demonstrations, and self-guided exploration.

II. EARNSHAW'S THEOREM AND THE THREE ESCAPE ROUTES

Module 0 places a free magnet above a fixed magnet. The user adjusts separation, field strength, and mass while a live $F(z)$ plot marks restoring (green) and unstable (red) bands. The floater starts in the green well; a *Nudge* command tips it onto the red hill approach, from which it slides off the saddle (Figs. 1 and 1). Parameter sliders drift the field gradually so students can explore the well before crossing the stability margin.

The three escape routes previewed are: (1) active feedback (Modules 1–2); (2) gyroscopic spin (Module 3); (3) driven rotation (Module 4).

III. THE FIVE MODULES AND IMPLEMENTATION

The application is built in React, TypeScript, and `three.js`. Physics integrates on a fixed 1 ms timestep decoupled from rendering; feedback modules use semi-implicit Euler; rigid-body modules use RK4. A startup self-check logs pass/fail diagnostics for each module preset.

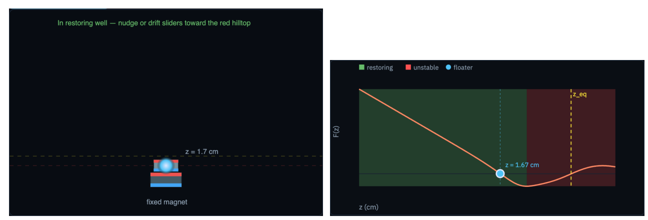

FIG. 1 (Left) Earnshaw primer: two magnets on a vertical axis. (Right) $F(z)$ with green restoring well, red unstable hill, and live floater marker; yellow line marks the hilltop z_{eq}.

III.A. One- and two-dimensional feedback

Module 1 models the floating-globe class: stable horizontally, unstable in z, corrected by PID control

$$I = \mathrm{clamp}\left(K_p e + K_i \int e\,dt + K_d \dot{e},\ 0,\ I_{\max}\right). \tag{1}$$

Live plots show gap error and coil current versus time; detuning K_p, temperature, or loop delay drives an eigenvalue across $\mathrm{Re}\,\lambda = 0$ and produces a visible crash.

Module 2 is the mathematically distinct sibling: stable in z, unstable in x and y, corrected by four horizontal coils with per-axis PID. Real-time traces display horizontal position and coil commands i_x, i_y as students lower gain, add coil imbalance, or increase loop delay (Fig. 2). Both feedback modules compute the closed-loop Jacobian and display eigenvalues live (Fig. 3).

Thermal drift with $K_i = 0$ produces visible steady-state sag below setpoint; raising K_i eliminates the offset — a teaching point tied to real levitating-globe trim pots.

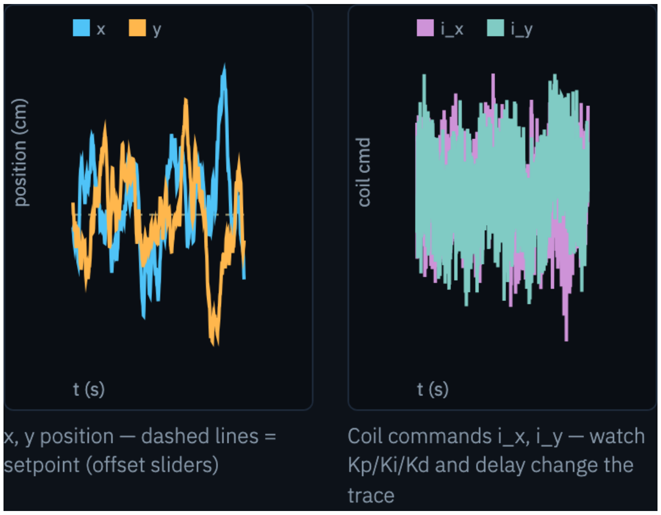

FIG. 2 Module 2: live horizontal position and coil-command traces while PID parameters are adjusted.

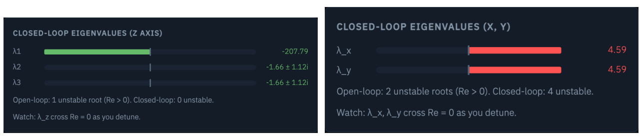

FIG. 3 Closed-loop eigenvalues for Module 1 (z instability tamed) and Module 2 (x, y instabilities tamed). Open-loop unstable roots cross $\mathrm{Re}\,\lambda = 0$ when detuned.

III.B. Dynamical rotor levitation

Module 4 integrates a floater in a motor-driven rotor field[5–7]. Trapping onset is governed by rotor geometry; the module displays $U_f(z_f)$ morphing as the lateral-displacement parameter crosses its critical value (Fig. 4).

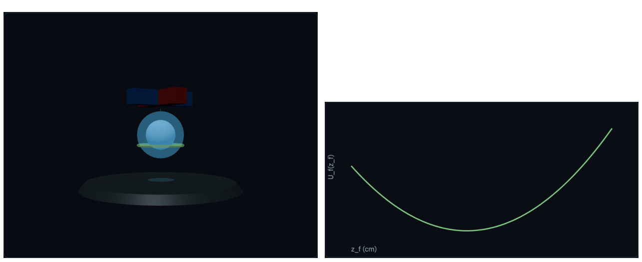

FIG. 4 Dynamical rotor module: frequency-locked floater (left) and floater potential $U_f(z_f)$ (right).

IV. THE GYROSCOPIC TOP AND THE 8.223 SYMMETRIC-TOP ANALYSIS

The classic Levitron is stabilized by angular momentum and magnetic force alone[3,4]. Module 3 implements the symmetric-top mechanics developed in the author's MIT 8.223 project[1,2].

IV.A. Lagrangian, conserved momenta, and precession roots

For a symmetric top with Euler angles (φ, θ, ψ) (precession, nutation, spin), the conserved momenta are

$$p_\psi = I_s(\dot\psi + \dot\varphi\cos\theta), \tag{2}$$

$$p_\varphi = (I_p\sin^2\theta + I_s\cos^2\theta)\dot\varphi + I_s\dot\psi\cos\theta. \tag{3}$$

At fixed θ, there are slow and fast precession roots; the slow root satisfies $\dot\varphi_{\mathrm{slow}} = mgl/p_\psi$. Launching at the slow-precession equilibrium with $\dot\theta = 0$ and $\cos\theta_{\mathrm{stable}} = p_\varphi/p_\psi$ is essential — arbitrary transverse angular velocity at $\theta = 0$ ejects the top immediately.

IV.B. Restoring potential and bounded nutation

Nutation satisfies an effective one-dimensional equation with potential

$$U_{\mathrm{res}}(\theta) = \frac{p_\psi^2}{2I_s} + \frac{(p_\varphi - p_\psi\cos\theta)^2}{2I_p\sin^2\theta} + V_{\mathrm{eff}}(\theta), \tag{4}$$

where V_{eff} combines gravity and magnetic upright bias (replacing $mgl\cos\theta$ in the gravitational-top form). Bounded nutation requires $\theta_{\min} < \theta < \theta_{\max}$ at the turning points of U_{res} (Fig. **??**). The device "works only when nice and upright": the allowable cone half-angle θ_c shrinks as spin drops toward $\omega_{\min}$ or as the field weakens with temperature.

The spin-rate window ($\omega_{\min} \approx 19$ rps for ferrite) and nutation cone are *both* required: the window ensures adiabatic following of the field (Berry phase[4]); the cone keeps the axis upright enough for that following to hold. Startup diagnostics verify $p_\psi^2 \gg 4mglI_p\cos\theta$ and that U_{res} possesses a local minimum before integration begins.

V. FROM TOY TO ATOM TRAP

A neutral atom with spin angular momentum is a microscopic gyroscope; magnetic trapping uses the same adiabatic-invariant logic as the Levitron[4,10]. The tool includes a dedicated page drawing this correspondence explicitly.

VI. CLASSROOM USE, AVAILABILITY, AND SUPPLEMENTARY MATERIAL

The Levitron Lab runs fully client-side with no backend. Instructors can run it locally (`npm install && npm run dev`), in GitHub Codespaces (Appendix A), or deploy a public instance on Railway (Appendix B). Figure capture for this manuscript uses `npm run figures` (Playwright headless screenshots of each module). Source code, data, and build scripts: https://github.com/saleemaldajani/levitron-lab.

VII. CONCLUSION

By rendering the levitating body alongside the equations that govern it, and by ordering devices according

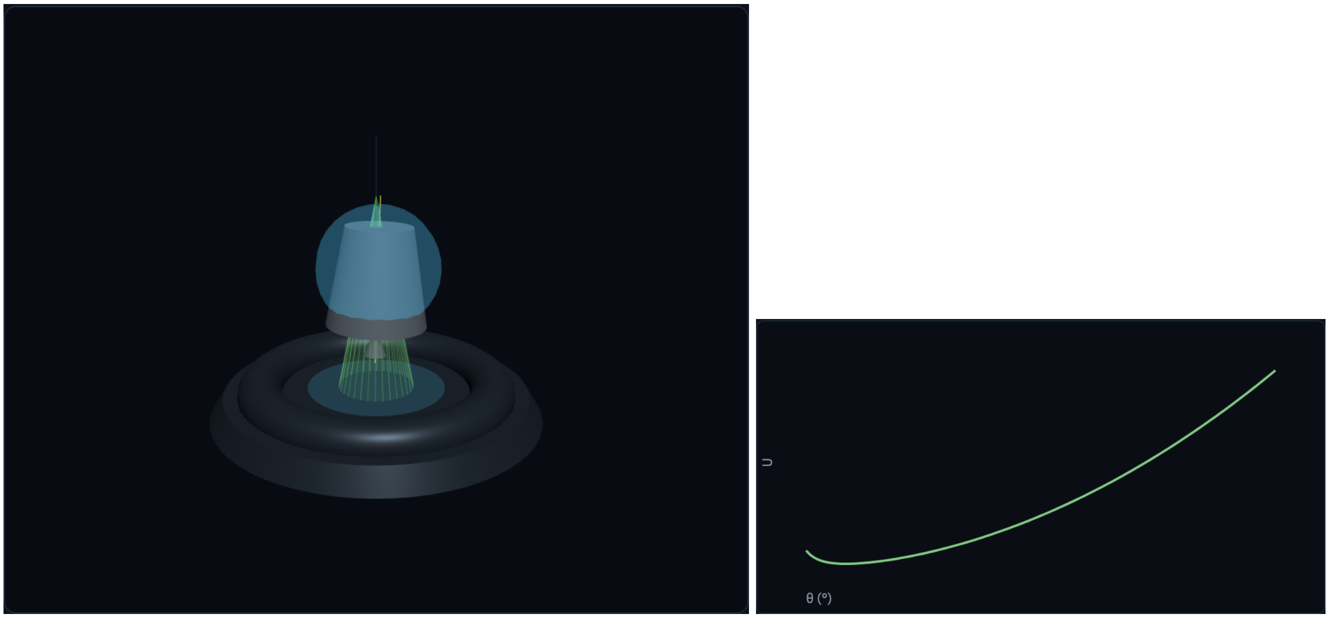

FIG. 5 (Left) Stable gyroscopic Levitron with nutation cone and symmetry axis. (Right) $U_{\text{res}}(\theta)$ showing the restoring well; $\theta_{\min} < \theta < \theta_{\max}$.

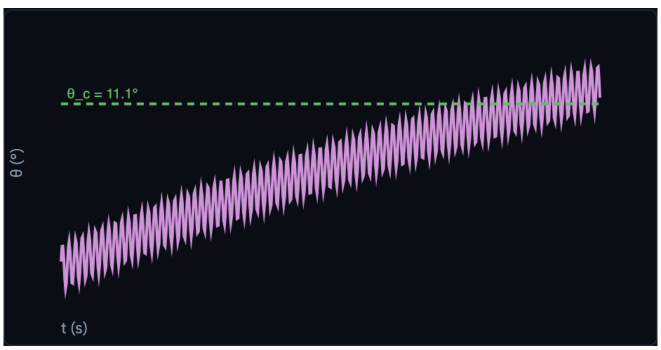

FIG. 6 Nutation angle $\theta(t)$: bounded nodding when stable; diverging growth before crash.

to unstable dimensionality, The Levitron Lab turns familiar toys into a unified quantitative curriculum from Earnshaw's prohibition to atom-trap analogies.

ACKNOWLEDGMENTS

The author thanks W. Ketterle for the conversation that motivated this work.

Appendix A: Running in GitHub Codespaces

The repository includes a `.devcontainer/devcontainer.json` configuration for one-click cloud development.

1. Open https://github.com/saleemaldajani/levitron-lab.
2. Select **Code** → **Codespaces** → **Create codespace on main**.
3. Wait for `npm install` (`postCreateCommand`) to finish; the dev server starts automatically (`postAttachCommand`: `npm run dev -- --host`).
4. In the **Ports** tab, open the forwarded preview for port **5173**.

Manual restart inside the codespace:

```
npm run dev -- --host
```

Production build and local preview:

```
npm run build
npm run preview -- --host
```

Regenerate paper figures (requires Playwright Chromium):

```
npx playwright install chromium
npm run figures
```

Appendix B: Deploying on Railway

The repository ships with `railway.toml`, `nixpacks.toml`, and an `npm start` script that serves the Vite production build with `serve`.

1. Create a project at https://railway.app and connect the GitHub repository `saleemaldajani/levitron-lab`.
2. Railway detects the Node build: `npm ci && npm run build`, then `npm start`.
3. Assign a public domain in the Railway dashboard; the health check probes `/`.

Local test of the production server:

```
npm run build
PORT=3000 npm start
```

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